

Muhammad Saad Imran

Machine Learning & AI Engineering · Retrieval Systems and Honest Model Validation

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PROFILE

Software Engineering student at Capital University of Science and Technology, Islamabad, Pakistan (CGPA 3.66) currently working as a Machine Learning Intern at FlyRank AI, building and validating ranking models on a real 81-million-row search dataset. My focus is the part of an AI system that decides whether there is enough evidence to answer at all — retrieval that abstains rather than guesses, and models validated with client-level holdouts so the reported number survives contact with data it has never seen. Comfortable in Python, pandas, scikit-learn and DuckDB, and equally comfortable reporting when my own assumption turned out to be wrong.

EXPERIENCE

Machine Learning Intern — FlyRank AI

Remote / International (Sarajevo, Bosnia and Herzegovina) · July 2026 – Present

- Building a click-through-rate opportunity model on FlyRank's 81-million-row search warehouse, querying it in place with **DuckDB over cloud-hosted Parquet** rather than downloading it, so the whole analysis runs inside a 12 GB notebook environment.
- Designed a **time-aware, client-grouped validation scheme** — features drawn from a prior window, the label from a later one, and GroupKFold on client so the model is never tested on a client or a period it has already seen. Verified zero client overlap in code.
- Model **beat the frozen rule baseline in every fold**: precision@50 rose from 0.492 to 0.784 against a 0.216 base rate. Reported alongside it that roughly half of that gain disappeared when click-derived features were removed (0.784 → 0.604).
- Ran a deliberate **leakage experiment** on real data: adding one label-derived column moved R^2 from an honest 0.037 to a flattering 0.874. Removed it and kept the honest number.
- **Tested a proposed ranking rule before building on it** and found it only 5.6% better than a flat prediction, with its top 100 pages tied at a single score. Rewrote the submitted analysis to report the real result and redesigned the task around an observed measurement.
- Designed and built a **document-grounded agent** that gathers evidence from a repository and file archive and marks every claim verified with its source or unverified with a question — using read-only tool restrictions so it structurally cannot write, publish or send.

PROJECTS

DecisAI — AI Bid-Response and RFP Decision Support System

CUST Hackathon 2026 · Team of three · My focus: the retrieval layer · React · FastAPI · Python · ChromaDB · Sentence Transformers

- Built **semantic retrieval** over a company's historical bid and capability records using Sentence Transformers and ChromaDB, so each requirement extracted from a new tender is matched against work the company has actually delivered before.
- Designed the system to mark a requirement “**No capability match**” when no supporting evidence exists, rather than drafting a plausible response — the behaviour that separates it from a generic writing tool, and the reason a proposal team can trust its output.
- Dual inference paths: **Claude Haiku 4.5** online and **Ollama phi3: mini** offline, so the system remains usable where tender documents cannot leave the network.
- Ran a **500-page synthetic tender end to end** — 91 requirements extracted at 94% confidence, each checked against retrieved evidence, with unsupported items flagged rather than hidden.

- **Identified and documented the system's limit:** retrieval degraded past roughly 1,500 pages. Chose to make it reliable at working sizes within the hackathon window instead of chasing scale, and recorded that as the first thing to fix.

Automated Testing Framework for an E-Commerce Admin Panel

Java · Selenium WebDriver · Selenium Grid · Docker · JUnit 5 · PHPUnit · Postman

- Designed a multi-layer test framework covering UI, unit, backend and API for a CRUD application.
- Ran **cross-browser tests in parallel** using Selenium Grid inside Docker containers.
- Combined Selenium WebDriver end-to-end cases, JUnit 5-unit tests, PHPUnit backend tests and Postman collections into a single regression suite.

Unbeatable Tic-Tac-Toe AI — Python · Pygame

- Implemented the **Minimax** algorithm with depth-based evaluation for provably optimal play, as an applied exercise in adversarial search and game-tree evaluation.

TECHNICAL SKILLS

Machine Learning & Data Python · pandas · NumPy · scikit-learn · DuckDB · Parquet · Jupyter / Google Colab · grouped & time-aware validation · permutation importance · leakage detection

AI Systems Retrieval-Augmented Generation (RAG) · Sentence Transformers · ChromaDB · prompt engineering · LLM evaluation · Model Context Protocol (MCP) · agent design and guardrails

Software Development JavaScript · Java · PHP · SQL · React.js · FastAPI · REST APIs · Tailwind CSS · HTML5 / CSS3 · MySQL / MariaDB

Testing & Tooling Selenium WebDriver & Grid · Docker · JUnit 5 · PHPUnit · Postman · TestNG · Maven · Git & GitHub · Jira (Agile)

EDUCATION

BS Software Engineering — Capital University of Science & Technology (CUST), Islamabad

Expected graduation 2027 · CGPA 3.66 / 4.00

Relevant coursework: Data Structures & Algorithms · Database Systems · Artificial Intelligence · Software Design & Architecture · Information Security · Automated Software Testing

CERTIFICATIONS

Anthropic Academy

AI Fluency: Framework & Foundations · Claude 101 · Introduction to Claude Cowork · AI Capabilities and Limitations · AI Fluency for Students · Introduction to Agent Skills (6 certificates, Jul–Aug 2026)

Google (Coursera)

Google Prompting Essentials · Google AI Essentials · Google AI: Fundamentals, Research & Insights, Brainstorming & Planning, Writing & Communicating (Jul–Aug 2026)

Atlassian (Coursera)

Agile with Atlassian Jira (Jul 2026)